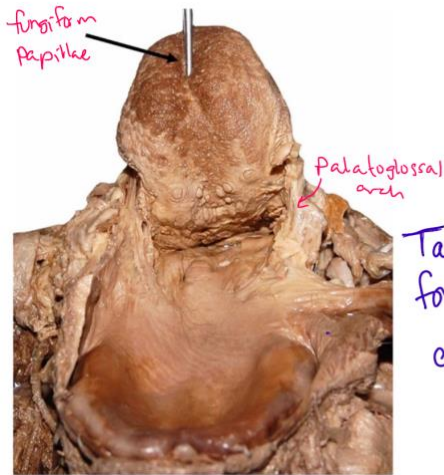


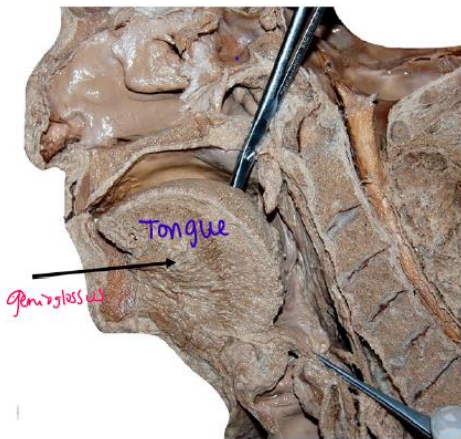
Term 2 Lab Exam Review



The cell bodies of the taste sensation fibers that innervate the tagged area are found in which of the following ganglion?

- A. Trigeminal sensory ganglion CN V
- B. Pterygopalatine parasympathetic ganglion - greater petrosal n. (branch of facial n.)
- C. Geniculate sensory ganglion CN VII
- D. Submandibular parasympathetic ganglion CN VII
- E. Glossopharyngeal

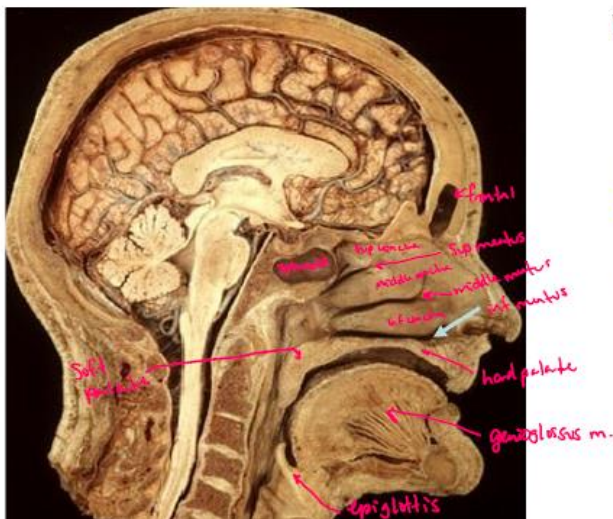
Taste sensation fibers for Ant 2/3 tongue
= Chorda Tympani (CN VII)
↳ Geniculate Ganglion



Which of the following nerves innervation the tagged structure?

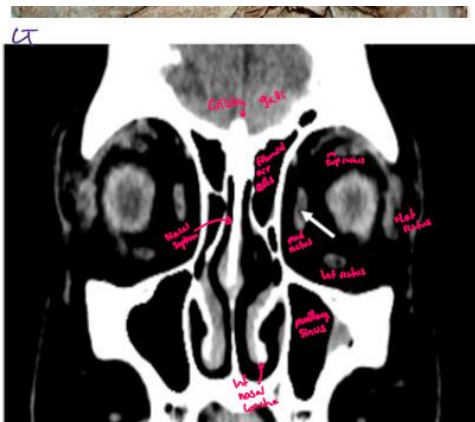
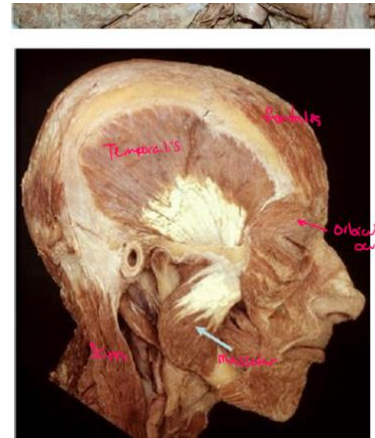
- A. Glossopharyngeal Stylopharyngeus
- B. Nerve to mylohyoid mylohyoid, ant belly digastric
- C. Vagus all pharynx except stylopharyngeus (IX), all palate except tensor veli palatini (V3)
- D. Hypoglossal all muscles of tongue except palatoglossus (CN X)
- E. Mandibular MAST MATT

Recurrent laryngeal (CN X) all larynx except cricothyroid (ant laryngeal (CN X))
Ansa cervicalis all infrahyoid except thyrohyoid (C1 which travels w/ hypoglossal)



Which of the following structures drains into the tagged areas?

- A. Maxillary sinus middle meatus
- B. Frontal sinus
- C. Sphenoid sinus Sphenoethmoidal recess
- D. Anterior ethmoidal air cells middle meatus
- E. Nasolacrimal duct Inf meatus
- F. Posterior ethmoidal air cells Superior meatus
- G. Middle ethmoidal air cells middle meatus

[illegible]

- A. Inferior alveolar nerve
- B. Inferior alveolar artery
- C. Nerve to mylohyoid
- D. Sphenopalatine artery
- E. Lingual nerve**
- F. Middle menigeal artery

Course in Skill
Course in Skill → provide sensation to lower teeth

foramen ovale $\rightarrow$ foramen
branch of maxillary a. (branch of ICA) $\rightarrow$ through IT fissura
from inf. alveolar bleeds mandibular foramen, runs in mylohyoid groove $\rightarrow$ supplies mylohyoid
branch of maxillary a., through sphenopalatine foramen into nasal cavity (back of superior meatus)
foramen ovale
foramen spinosum

maxillary n. (V₂) foramen rotundum

Tell Ziggy Bob Maly ^{Called}
anterior, buccal, marginal mandibular, cervical

- A. Facial branches to muscles of facial expression: Temporal, zygomatic, buccal, mental
- B. Mandibular MAST MAT
- C. Great auricular cutaneous: Skin of auricle, over parotid & mastoid process; also deep layer of parotid fascia
- D. Maxillary sensory (V2) to pterygopalatine foramen, nose, upper jaw teeth, lat. temporal
- E. Hypoglossal muscle of tongue except palatoglossus (CN X)

A. Internal carotid $\rightarrow$ MCA & ACA

- B. Common carotid ICA & ECA
- C. Superficial temporal frontal (ant) branch & parietal (post) branch
- D. External carotid maxillary a & sup. temporal a.
- E. Subclavian axillary a.

A. CN II afferent branch of pupil reflex

- A. CN V
- B. CN III SR, MR, IR, IO, efferent branch of pupal reflex
- C. CN IV SO
- D. CN VI LR
- E. CN VII efferent branch corneal reflex & lacrimation reflex

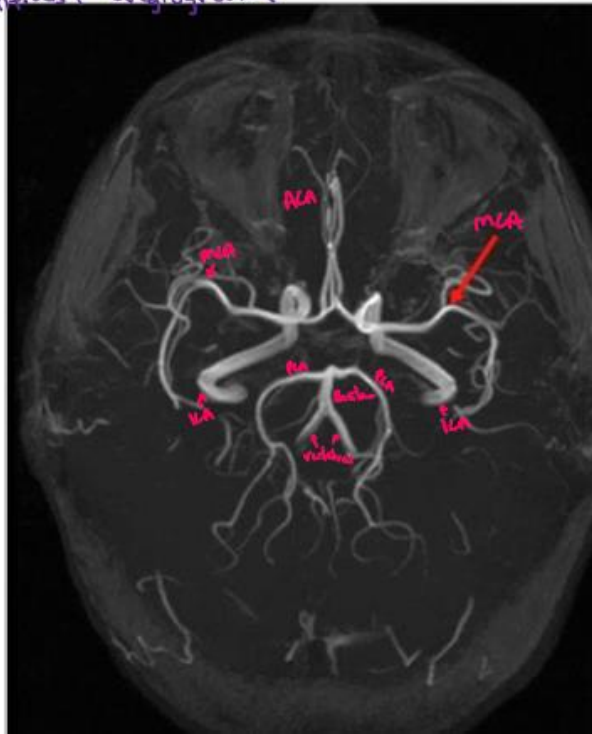
CN V, afferent branch corneal reflex & lacrimation reflex

CN V3 afferent & efferent jaw jerk

CN IX afferent gag

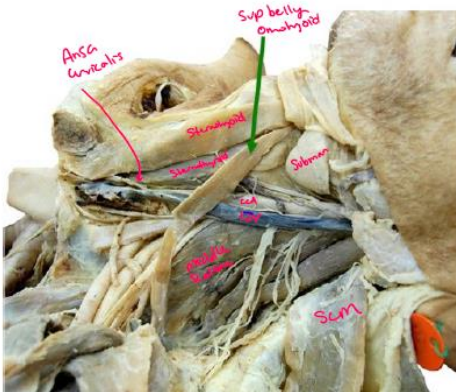
CN X efferent gag; afferent & efferent cough reflex

Contrast angiogram



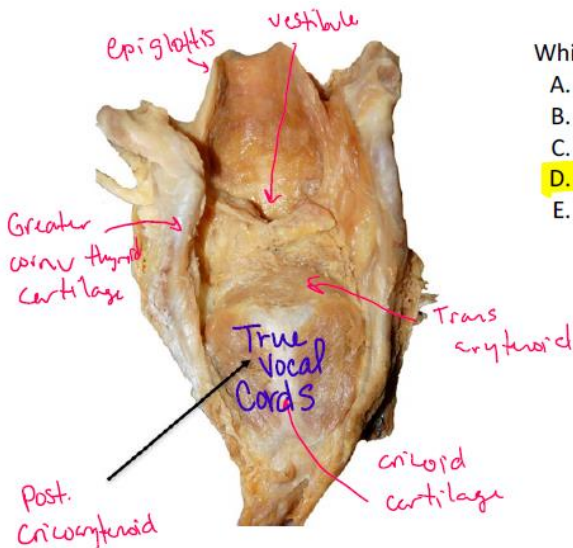
Identify the artery

- A. Internal carotid
- B. Anterior cerebral
- C. Posterior cerebral
- D. Superior cerebellar
- E. Middle cerebral**
- F. Anterior inferior cerebellar
- G. Basilar



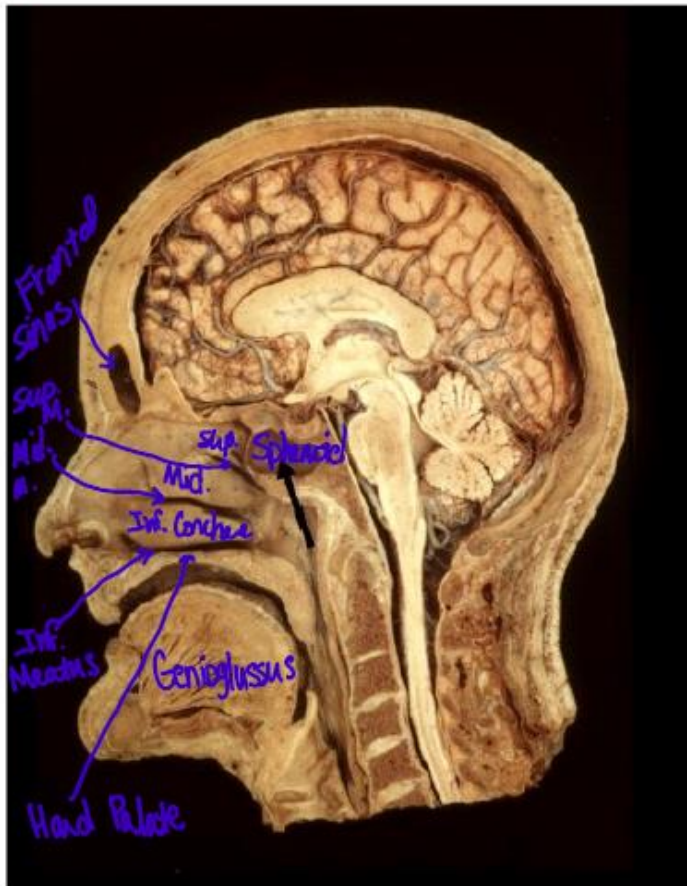
Which of the following nerves innervates the tagged muscle?

- A. Ansa cervicalis** *all infrahyoid muscles except thyrohyoid (C1 travels w/ hypoglossal)*
- B. Nerve to mylohyoid *mylohyoid, ant belly digastric*
- C. Great auricular *sensory to skin over parotid/auricle*
- D. C1 *thyrohyoid*
- E. Hypoglossal *all muscles of tongue except palatoglossus (CN X)*



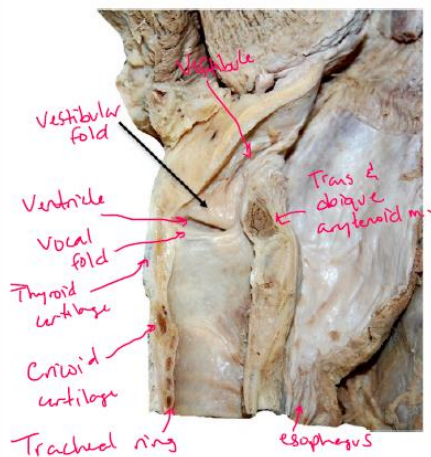
Which of the following nerves innervates the tagged muscle?

- A. Internal laryngeal *sensory innervation above vocal cords*
- B. Glossopharyngeal *stylopharyngeus m.*
- C. External laryngeal *cricothyroid m.*
- D. Recurrent laryngeal** *all intrinsic muscle of larynx except cricothyroid*
- E. Spinal accessory *scm & trapezius*



Identify the tagged structure

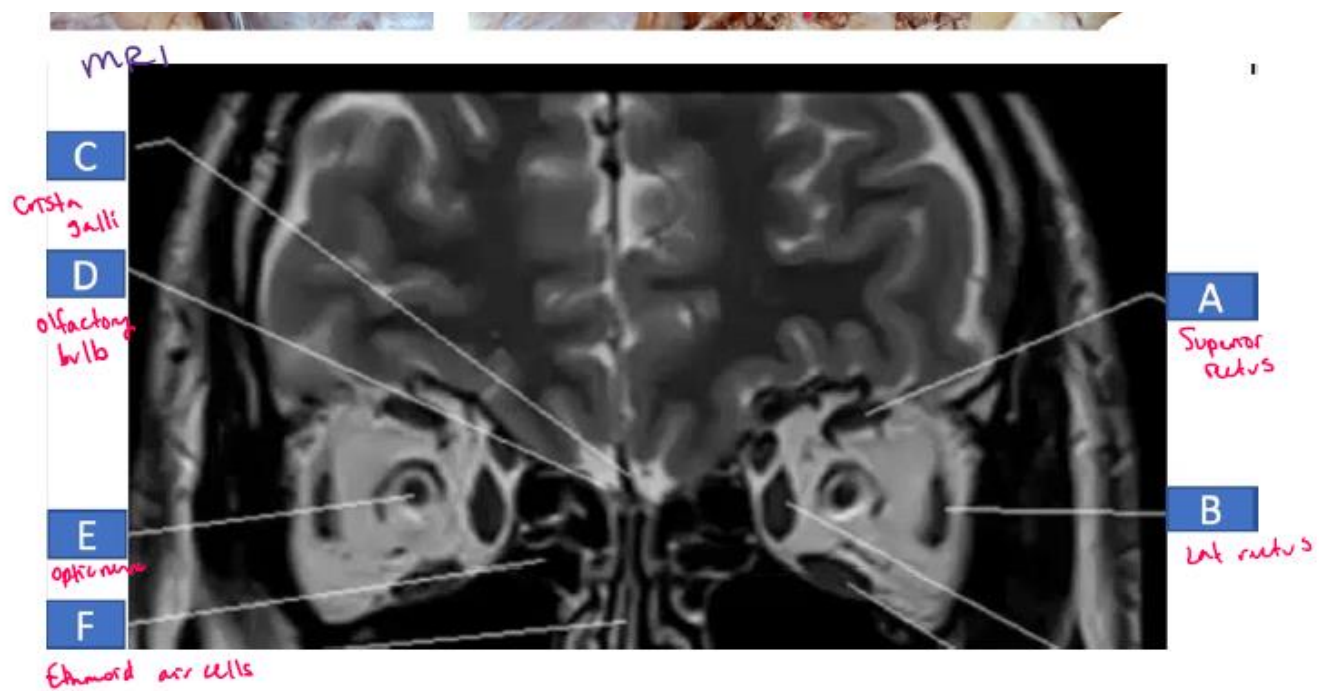
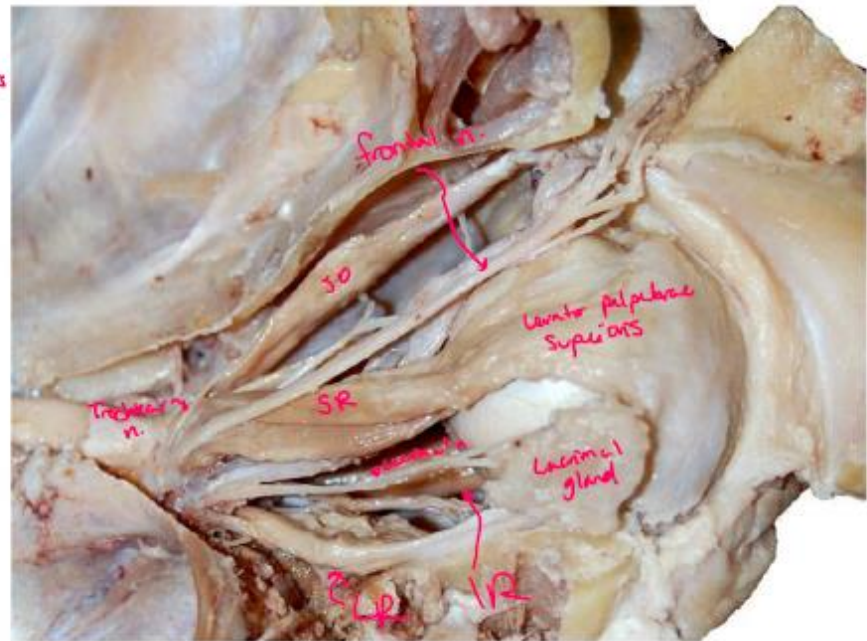
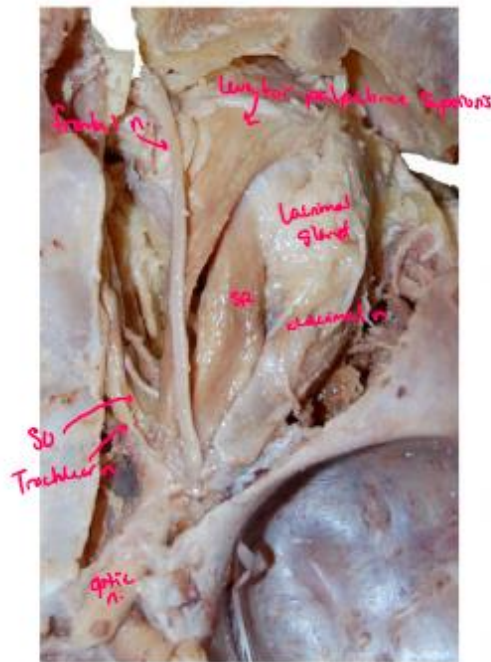
- A. Maxillary sinus
- B. Sphenoid sinus**
- C. Frontal sinus
- D. Ethmoid air cells



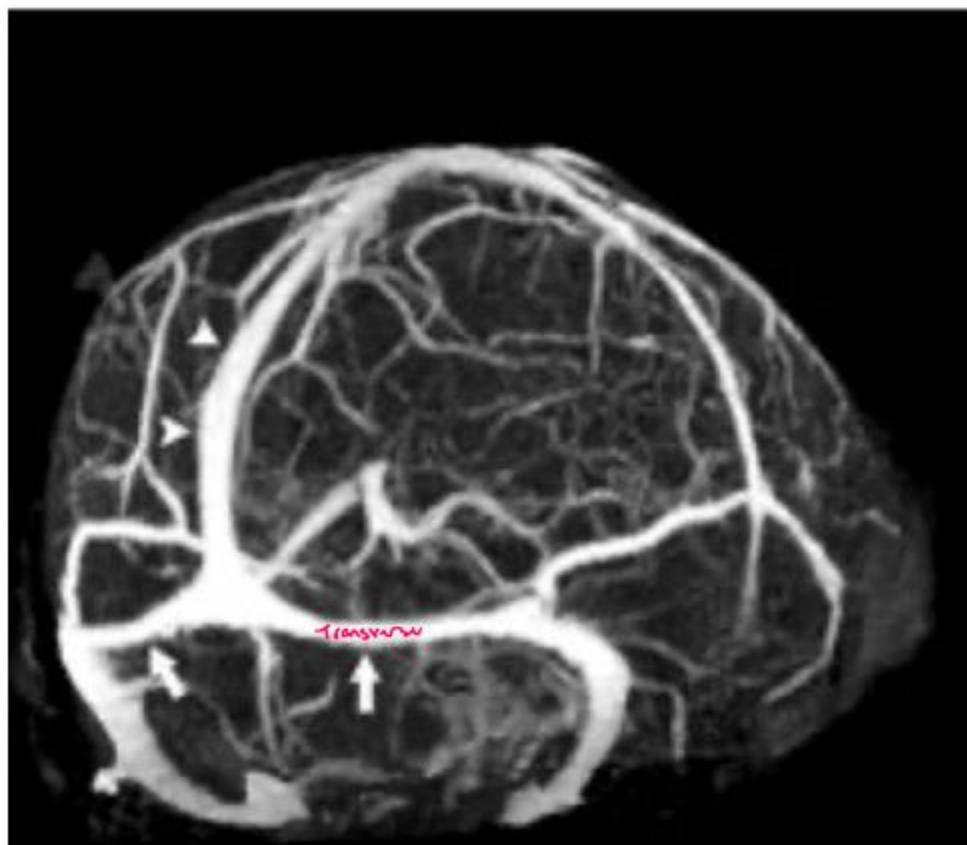
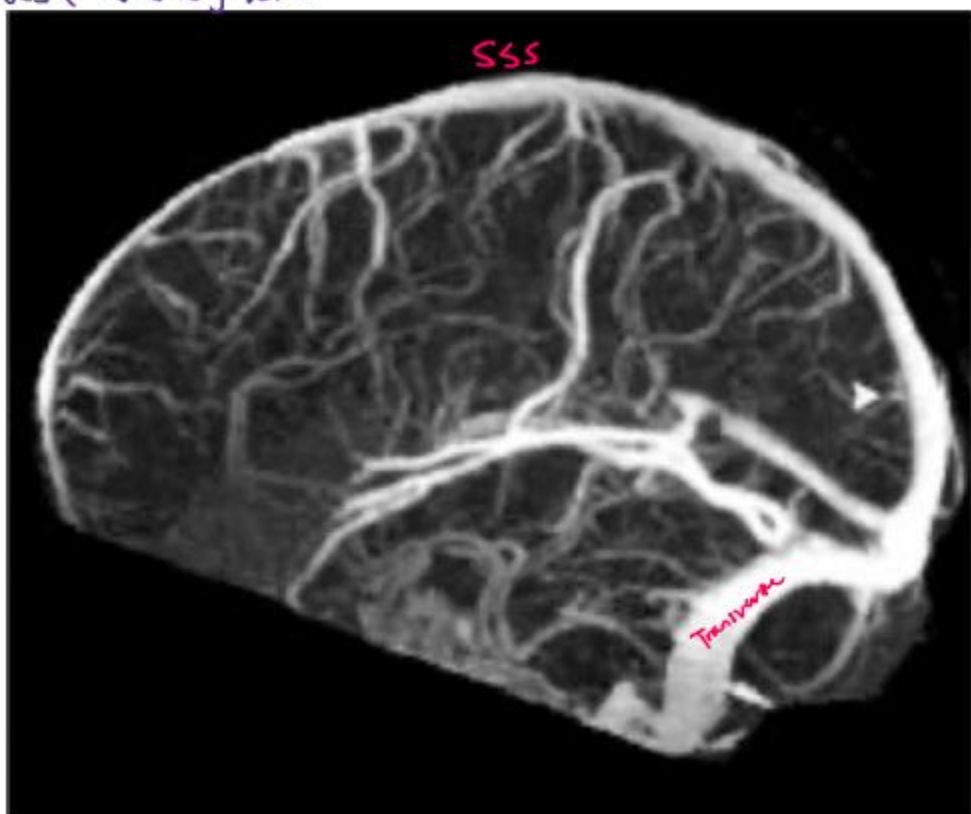
Which of the following nerves innervates the tagged area?

- A. Pharyngeal plexus motor & sensory supply to muscle of pharynx & soft palate
- B. External laryngeal cricothyroid m.
- C. Internal laryngeal** above vocal cord sensory
- D. Superior laryngeal branch of vagus & supplies larynx; gives off ext. laryngeal & int. laryngeal
- E. Recurrent laryngeal below vocal cord sensory/motor

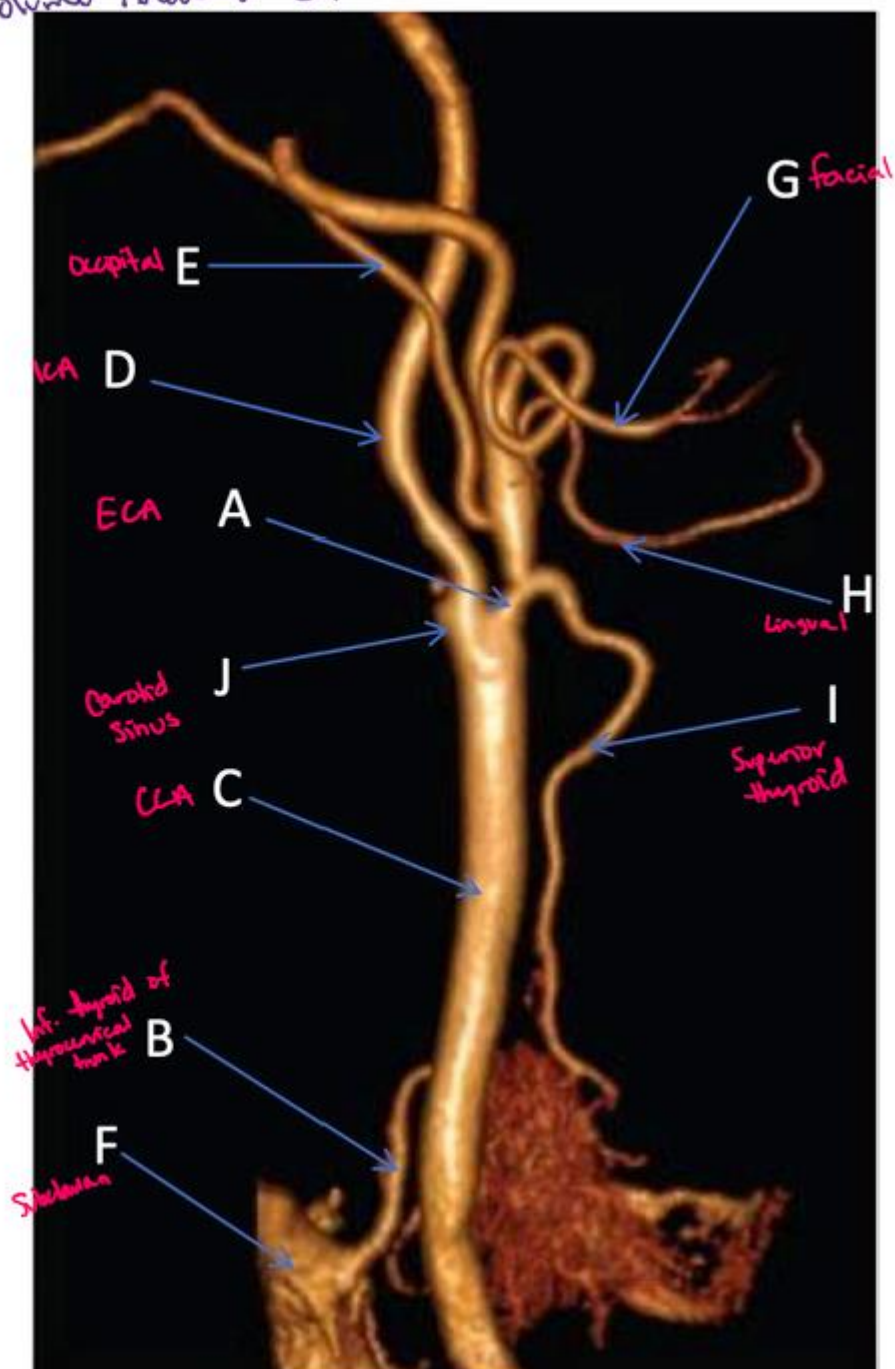
Term 2 Lab Exam Review

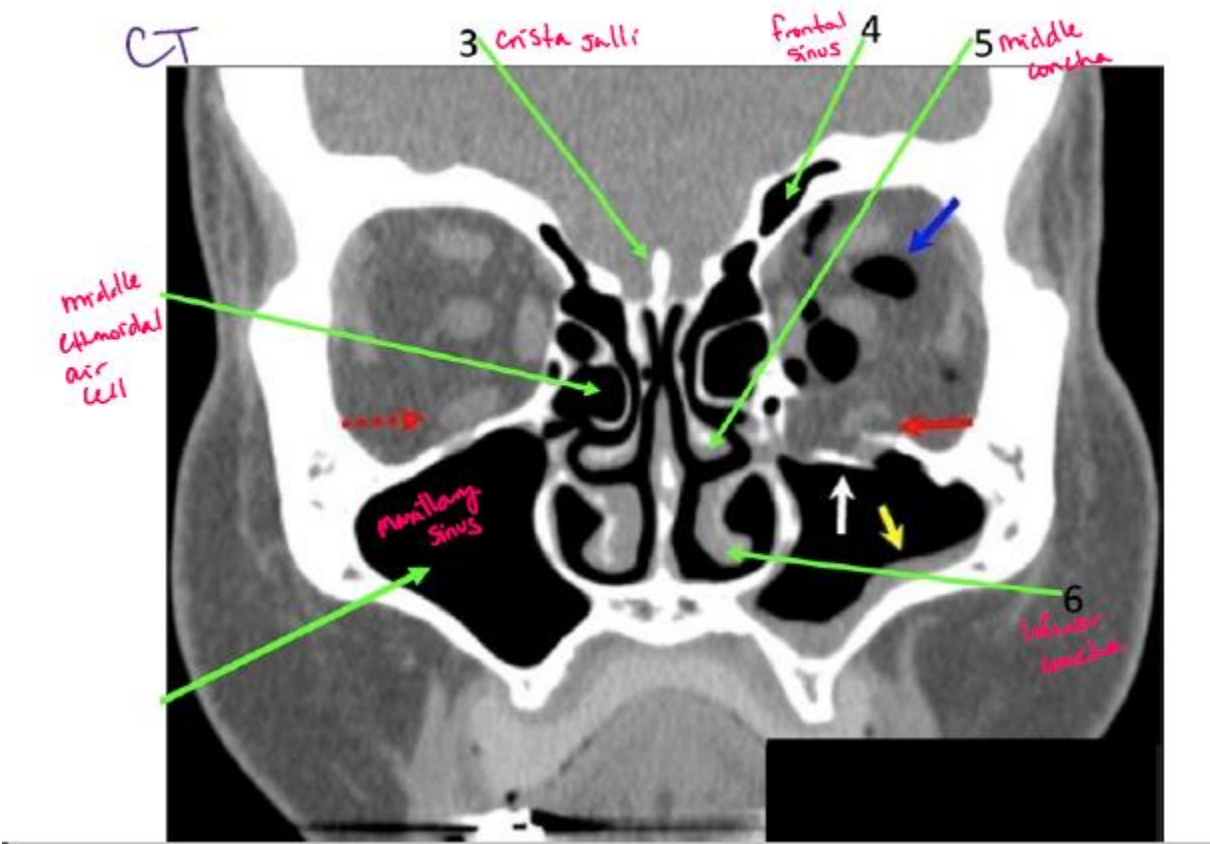


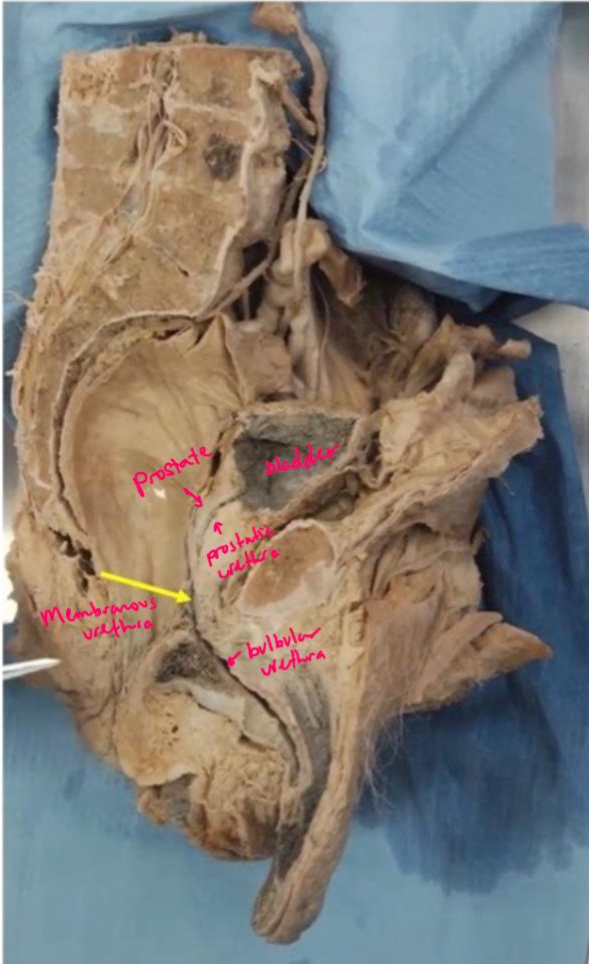
contrast venogram



volume rendered CT







Which of the following best describes the location of the part of the urethra at the tip of the arrow?

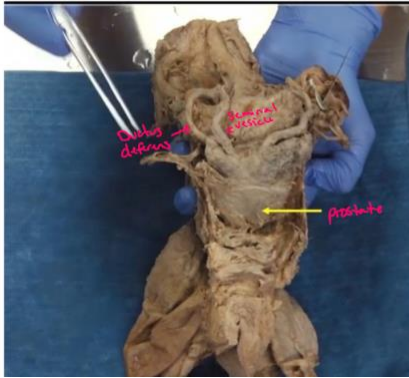
- A. In the prostate
- B. In the bladder
- C. In the penis
- D. In the superficial perineal pouch
- E. In the deep perineal pouch



Lymph from the structure at the tip of the arrow
Initially drains into which of the following areas?

- A. Superficial inguinal scrotum/skin, labia majora, lower 1/3 vagina, anus (below pectinate)
- B. External iliac body of uterus
- C. Internal iliac cervix, 2/3 vagina, uterus, prostate, bladder, inf rectum (above pectinate)
- D. Deep inguinal glans penis, glans clitoris, labia minora
- E. Common iliac ext. & internal lymph drains here

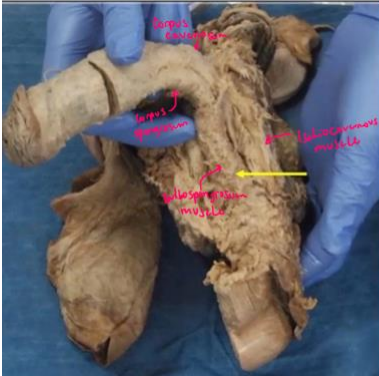
Term 2 Lab Exam Review



Which of the following arteries is the major source

Of blood supply to the structure at the tip of the arrow?

- A. Inferior vesical Supply bladder neck & prostate; branch of int. iliac a.
- B. External iliac Supply leg; branch of common iliac
- C. Testicular Supply testes; branch of abd. aorta
- D. Inferior rectal Supply rectum; branch of internal pudendal
- E. Internal pudendal Supply ext genitalia; branch of internal iliac



Which of the following nerves provides motor

innervation to the muscle at the tip of the arrow?

- A. Pelvic splanchnic parasympathetic (Uterus & upper 3/4 vagina, erection)
- B. Sacral splanchnic sympathetic (uterus, erection)
- C. Pudendal ext anal sphincter, ischioanal fossa, bulbospongiosus, muscles of perineal pouch
- D. Lumbar splanchnic Smooth muscle & glands of pelvic viscera & hindgut



Which of the following statements correctly describes

The structure at the tip of the arrow?

- A. It attaches the ovary to the uterus proper ovarian ligament
- B. It contains the ovarian vessels Suspensory ligament
- C. It attaches the uterus to the labia Majora round ligament
- D. It contains the uterine vessels broad ligament
- E. It is a remnant of the gubernaculum ovarium



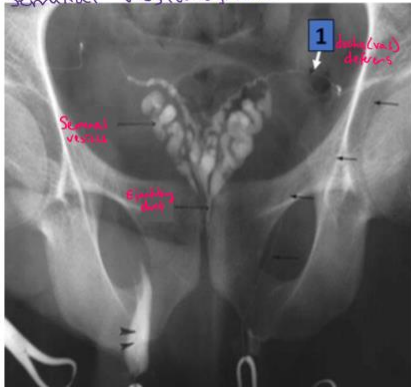
Which of the following nerve(s) provides the

sympathetic innervation to the structure at the tip of the arrow?

- A. Sacral splanchnic uterus
- B. Pelvic splanchnic parasympathetic
- C. Pudendal sweat glands/sm. muscle in skin & vessels in scrotum/labia, penis, anal region
- D. Ilioinguinal ant scrotum/labia majora, root of penis, upper antero-medial thigh

Term 2 Lab Exam Review

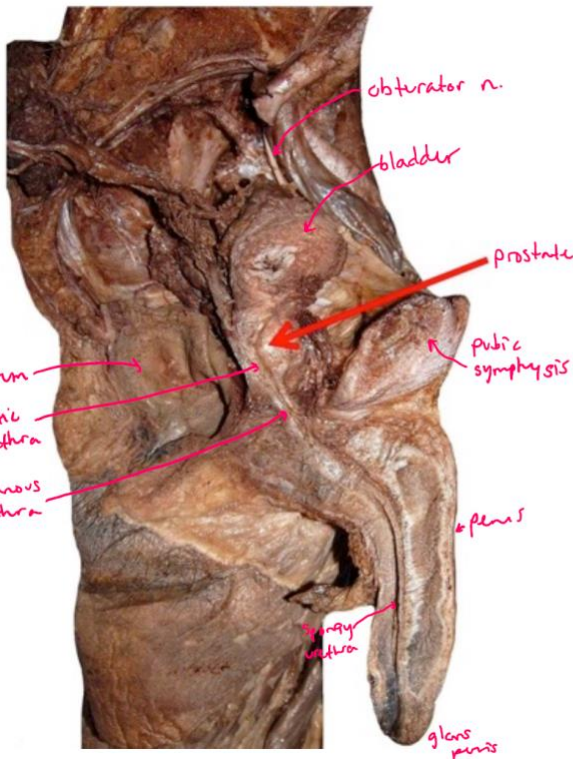
Seminal vesiculogram



A 37 year old man comes to the physician because him and his wife has been unable to have a child. Fertility tests show reduced semen volume and low sperm count. Ultrasound is inconclusive and a seminovesiculogram is performed which is shown. Which of the following is the embryological origin of the structure labeled 1?

- A. Paramesonephric duct *ovine tubes, uterus*
 B. Genital tubercle *penis/clitoris*
 C. Mesonephric duct *trigone of bladder, epididymis, vas deferens, ejaculatory duct, seminal vesicles*
 D. Urogenital sinus *urethra, bladder, prostatic urethra, membranous urethra, urethra, vagina*

Labioscrotal swellings *scrotum, labia majora*
 Urethral folds *ventral aspect of penis, spongy urethra, labia minora*



Which of the following provides parasympathetic innervation to the structure at the tip of the arrow?

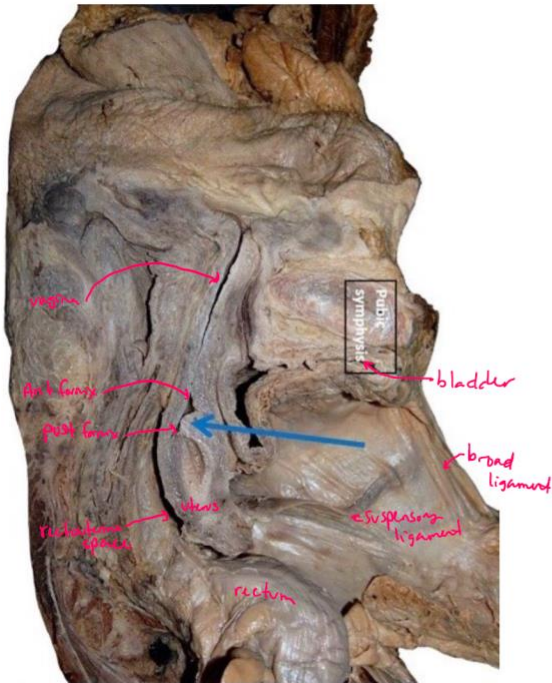
- A. Sacral splanchnic *Sympathetic*
 B. Pelvic splanchnic *Para-sympathetic*
 C. Pudendal *Somatic*
 D. Lumbar splanchnic



A 32 year old woman undergoes fertility assessment as she has been unable to become pregnant for the past 4 years. Fertility testing of her husband shows no problems. Hormone testing and ultrasound are normal. The patient undergoes a hysterosalpingogram. Which of the following best describes the location of the fluid as indicated by the circle?

- A. Uterine cavity
 B. Rectouterine pouch
 C. Uterine tube lumen
 D. Bladder
 E. Urethra
 F. Retropubic space

Term 2 Lab Exam Review



Pain fibers from the labeled structures travels back to the spinal cord with which of the following nerves?

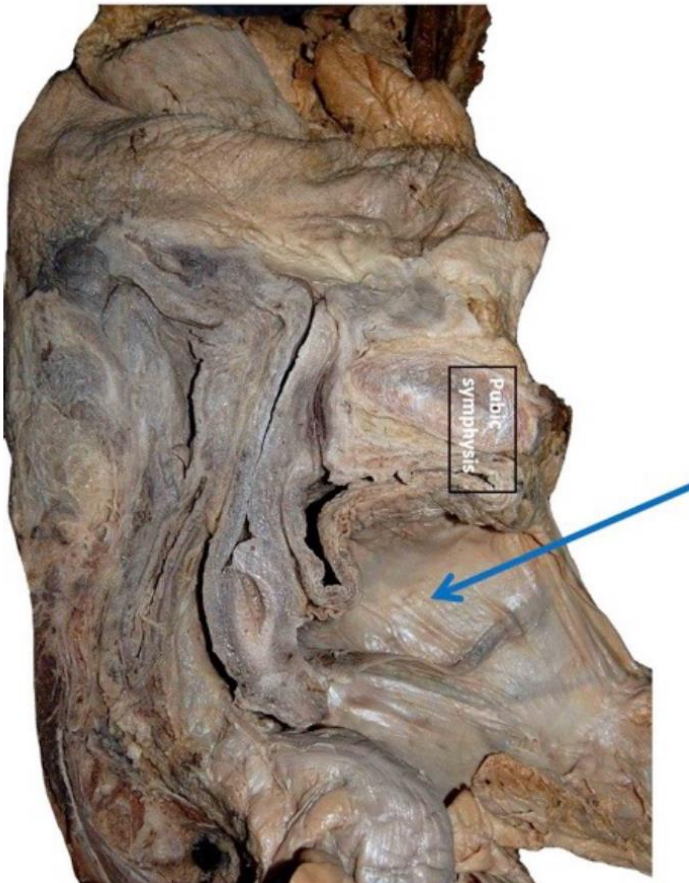
- A. Pelvic splanchnic *cervix*
- B. Lumbar splanchnic
- C. Sacral splanchnic *uterus*
- D. Pudendal Somatic *labia, lower 1/4 vagina, penis/clitoris*

Above pelvic pain line

- visceral afferents travel w/ sympathetics
- pain → T10-L1 areas
- uterine tubes, uterine fundus/body

Below pelvic pain line

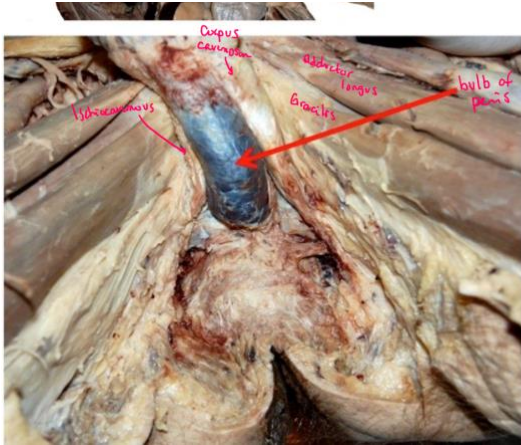
- visceral afferents travel w/ parasympathetics
- pain → S2-S4
- cervix, urinary bladder, upper vagina



Identify the ligament

- A. Suspensory ligament of ovary
- B. Uterus a real
- C. Ovarian
- D. Broad
- E. Round

Term 2 Lab Exam Review



Which of the following arteries provides blood supply to the structure at the tip of the arrow?

- A. Deep artery of the penis *corpus cavernosum; branch int. pudendal*
- B. Dorsal artery of the penis *glans penis, skin, fascia; branch int. pudendal*
- C. Artery of the bulb of the penis
- D. Testicular artery *testes; branch abdominal aorta*



Match the correct structures to the numbers in the image.

- 3 A. Membranous urethra
- 4 B. Spongy urethra
- 5 C. Corpora cavernosa
- 1 D. Bladder
- 2 E. Prostatic urethra

Term 2 Lab Exam Review



Match the correct structures to the numbers in the image.

- 4 A. External os of the cervix
- 5 B. Vaginal canal
- 2 C. Body of the uterus
- 7 D. Posterior fornix of the vagina
- 6 E. Anterior fornix of the vagina
- 3 F. Cervical canal
- 1 G. Fundus of the uterus



Match the correct structures to the numbers in the image.

- 1 A. Bladder
- 2 B. Uterus
- 3 C. Rectum

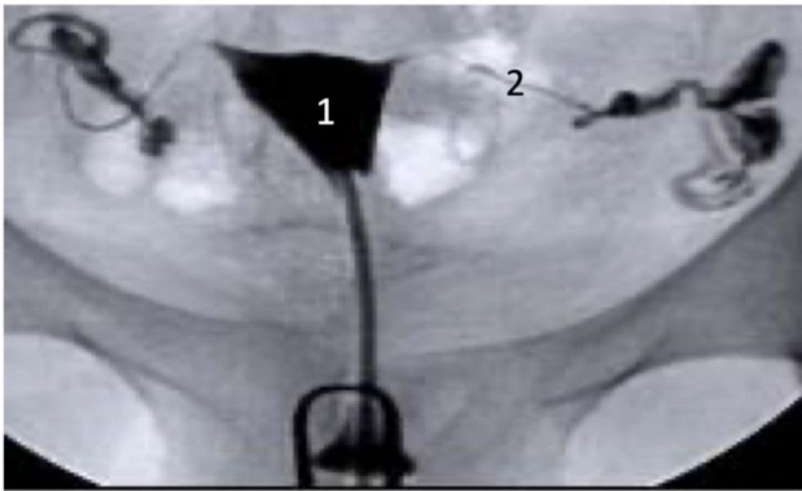
Term 2 Lab Exam Review

Nerve Supply

- **PUDENDAL NERVE—SOMATIC S2-S4**
 - External anal sphincter—inferior rectal nerve
 - Erectile tissue muscles
 - Muscles of deep perineal pouch
 - Anus
 - Labia
 - Scrotum
 - Penis
 - Clitoris
 - Lower 1/4 of vagina
 - Perineal nerve
 - Posterior labial
 - Dorsal nerve of Clitoris/Peris
 - **Pelvic Splanchnics → PSNS S2-S4**
 - Cervix
 - Upper 1/4 of Vagina
 - "Point" erection
 - **Sacral Splanchnics → SNS Hypogastric**
 - Uterus
 - "Shoot" ejaculation
- *IMPORTANT TAKE AWAY***
IN A PUDENDAL BLOCK—THE CERVIX AND UTERUS ARE NOT INCLUDED

LYMPHATICS

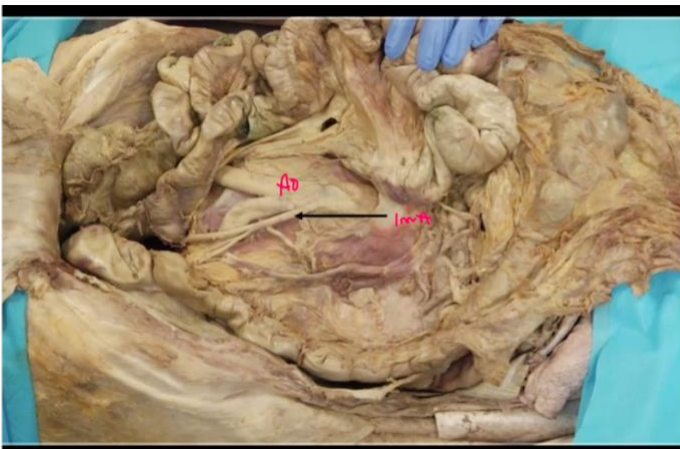
- **Superficial Inguinal:**
 - part of uterus where round ligament attaches
 - Scrotum (SKIN)***
 - Skin of Penis
 - Vestibule
 - Labia Majora
 - Lower 1/4 of Vagina
 - Anus: below pectinate line
- **Deep Inguinal:**
 - Glans Penis
 - Glans Clitoris
 - Labia Minora
- **Internal Iliac**
 - Inferior rectum above pectinate line
 - Cervix
 - Middle/Upper 1/2 Vagina
 - Uterus
 - Prostate
 - Bladder
- **External Iliac**
 - Body of the Uterus
- **Lumbar/Para-aortic**
 - Fundus of the Uterus
 - Ovaries
 - Fallopian Tubes
 - Testes



Hysterosalpingogram

1. Uterine body
2. Isthmus of Fallopian tube

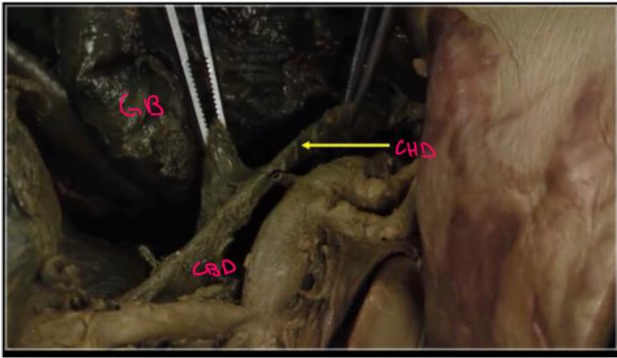
DM



The tagged vessel provides direct arterial supply to
Which of the following?

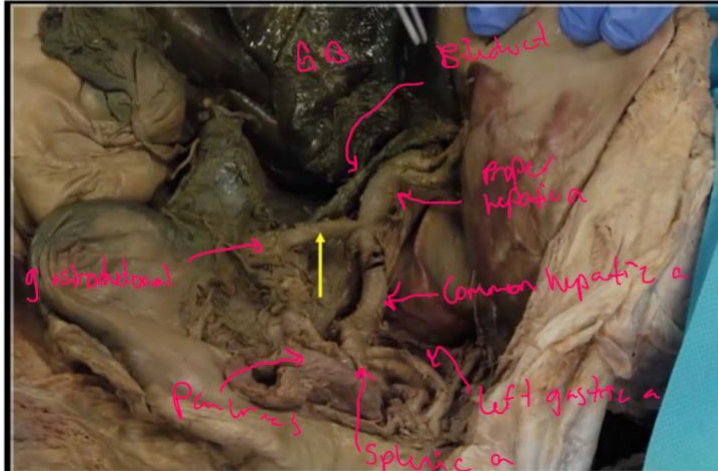
- A. Descending colon IMA
- B. Ascending colon SMA
- C. Transverse colon SMA
- D. Cecum SMA
- E. Right colic flexure SMA

Term 2 Lab Exam Review



Which of the following statements best describe The tagged structure?

- A. Blockage will not result in jaundice
- B. It carries bile from the liver *CHD*
- C. It carries bile from the gallbladder *cystic duct*
- D. It joins the main pancreatic duct to form the Hepatopancreatic ampulla *CBD*
- E. It is found in the hepatoduodenal ligament *portal triad*



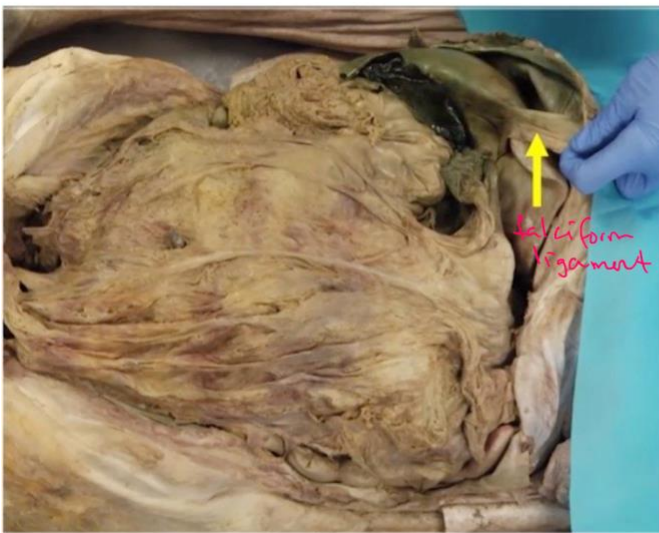
The tagged artery supplies which of the following Organs either directly or through its branches?

- A. Stomach and liver
- B. Liver and gallbladder
- C. Stomach and spleen
- D. Stomach and duodenum
- E. Liver and spleen



Which of the following statements best describes The peritoneal relations of the tagged structure?

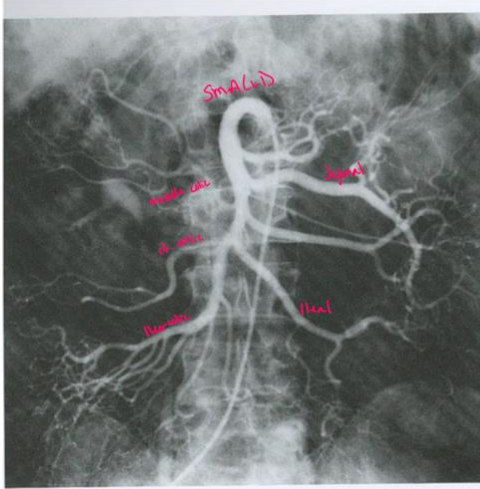
- A. It is primarily retroperitoneal *Kidneys, ureters, AO/IVC, sym. trunks*
- B. It is intraperitoneal *Stomach, liver, 1st duodenum, jejunum, ileum, spleen, cecum, appendix, trans & sigmoid colon*
- C. It is secondary retroperitoneal *Pancreas, duodenum (2nd, 3rd, 4th), Asc & desc. colon, rectum*
- D. It is made of adventitious



The tagged structure attaches the liver to Which of the following?

- A. Anterior abdominal wall *falciform ligament*
- B. Stomach *hepatogastric*
- C. First part of the duodenum *hepatoduodenal*
- D. Right kidney *hepatorenal*
- E. Transverse colon

Term 2 Lab Exam Review



This artery primarily gives blood supply to which of the following embryological areas?

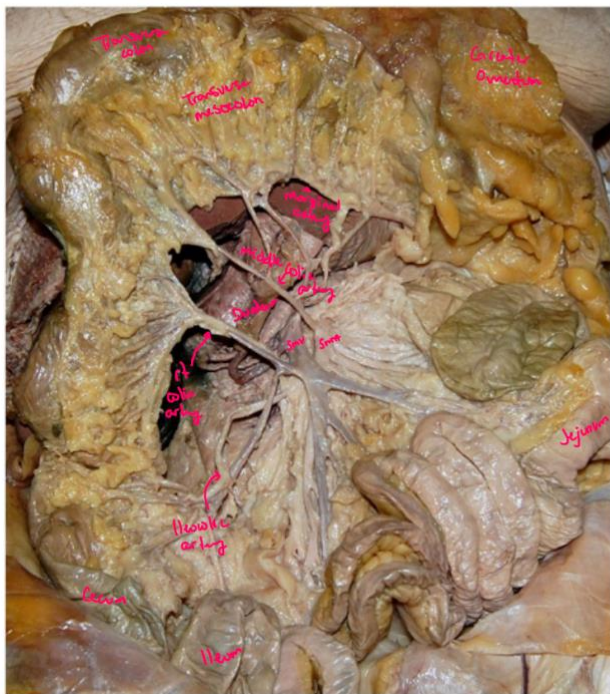
- A. Foregut *celiac*
- B. Midgut *SMA*
- C. Hindgut *IMA*
- D. Proctodeum below pectinate line - inf rectal a.

Above pectinate

- Sup rectal a
- Internal v. plexus → sup rectal v (portal)
- Internal iliac lymph
- Sympathetics - visceral afferents
- from hindgut (endoderm)

Below pectinate

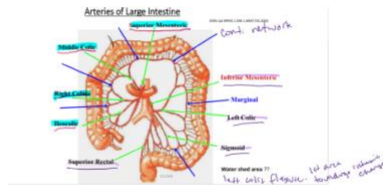
- Inf rectal a
- Ext v. plexus → inf rectal veins (portal)
- Sup. horizontal inguinal lymph
- Somatic afferents
- Proctodeum (ectoderm)



The tagged structure forms an anastomoses between the vessels that supplies which of the following embryological areas?

- A. Foregut and midgut
- B. Midgut and hindgut
- C. Midgut and proctodeum
- D. Hindgut and proctodeum
- E. Foregut and hindgut

water shed area
↳ connects SMA & IMA



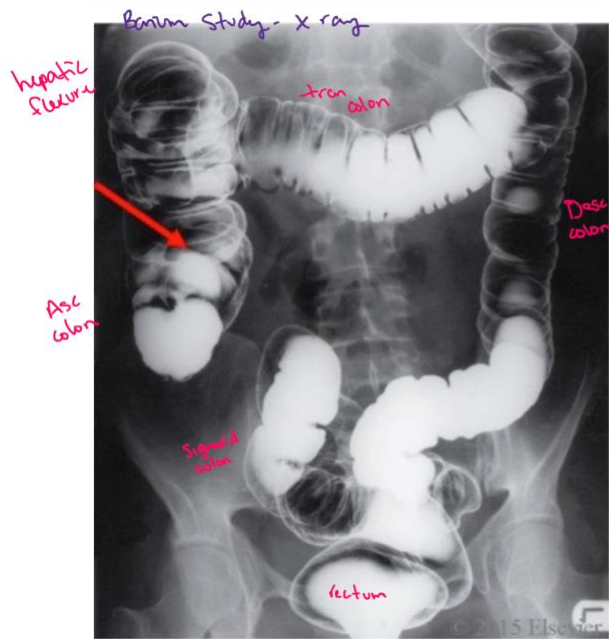
Endoscope



Which of the following portal caval anastomoses is responsible for the pathology seen in this image?

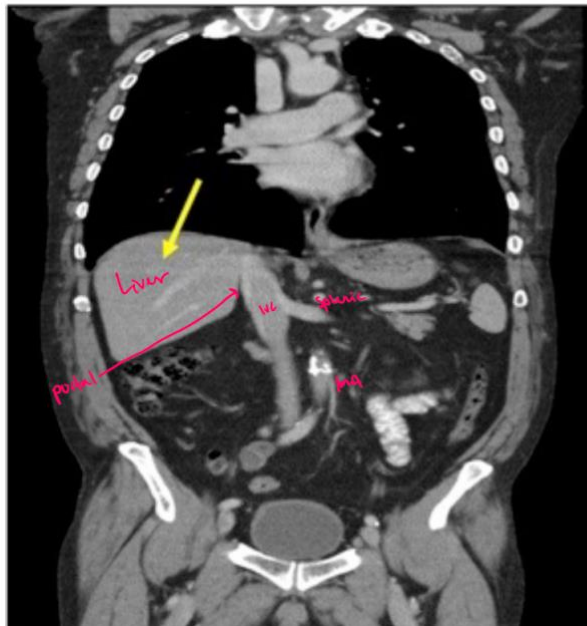
- A. Superficial epicanthic and paraumbilical *caput medusae*
- B. Middle and inferior rectal w/ superior rectal → hemorrhoids
- C. Esophageal and left gastric *hematemesis*
- D. Splenic and left colic

Term 2 Lab Exam Review



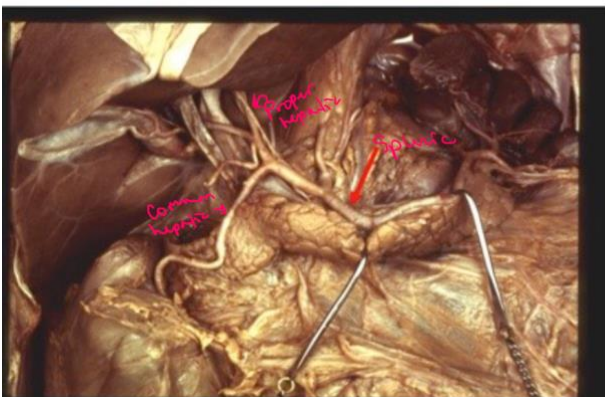
Which of the following gives parasympathetic innervation To this organ?

- A. Pelvic splanchnic hindgut
- B. Vagus foregut/midgut
- C. Lumbar splanchnic sympathetic - hindgut
- D. Greater thoracic splanchnic sympathetic - foregut
- E. Lesser thoracic splanchnic sympathetic - midgut
- F. Least thoracic splanchnic sympathetic - midgut



Visceral afferent fibers from the tagged organ will travel back To the spinal cord with which of the following nerves?

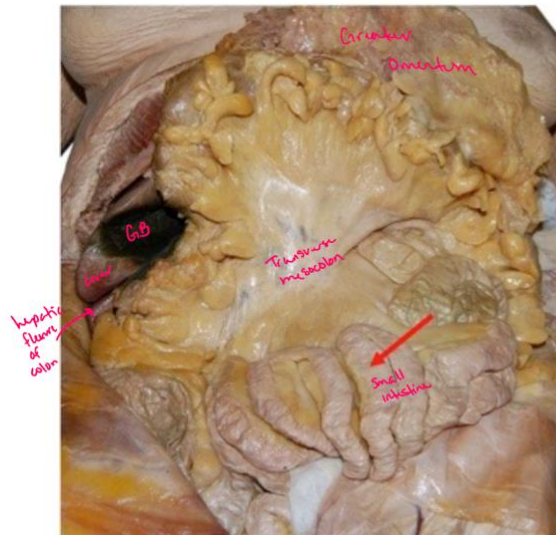
- A. Greater thoracic splanchnic sympathetic - foregut
- B. Lumbar splanchnic sympathetic - hindgut
- C. Lesser thoracic splanchnic sympathetic - midgut
- D. Least thoracic splanchnic sympathetic - midgut
- E. Pelvic splanchnic parasympathetic - hindgut
- F. Vagus parasympathetic - foregut & midgut



Which of the following is a branch of the tagged artery?

- A. Right gastro-omental gastroduodenal branch hepatic a.
- B. Right gastric hepatic artery
- C. Left gastric celiac trunk
- D. Left gastro-omental splenic a.
- E. Gastroduodenal hepatic a.

Term 2 Lab Exam Review



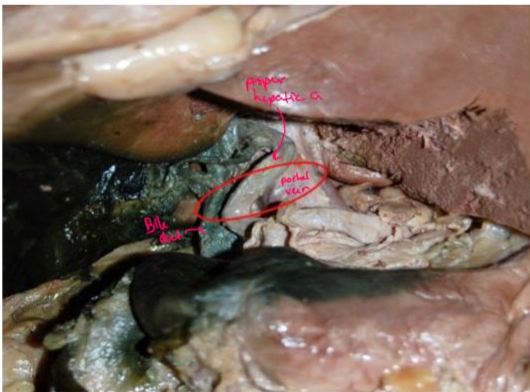
Which of the following groups of lymph nodes drain the labeled structure?

- A. Inferior mesenteric *hind gut*
- B. Celiac *fore gut*
- C. Lumbar *testes/ovaries*
- D. Internal iliac *cervix, upper 2/3 vagina, prostate, bladder, uterus*
- E. Superior mesenteric *mid gut*



Match the correct name to the structures numbered in the image.

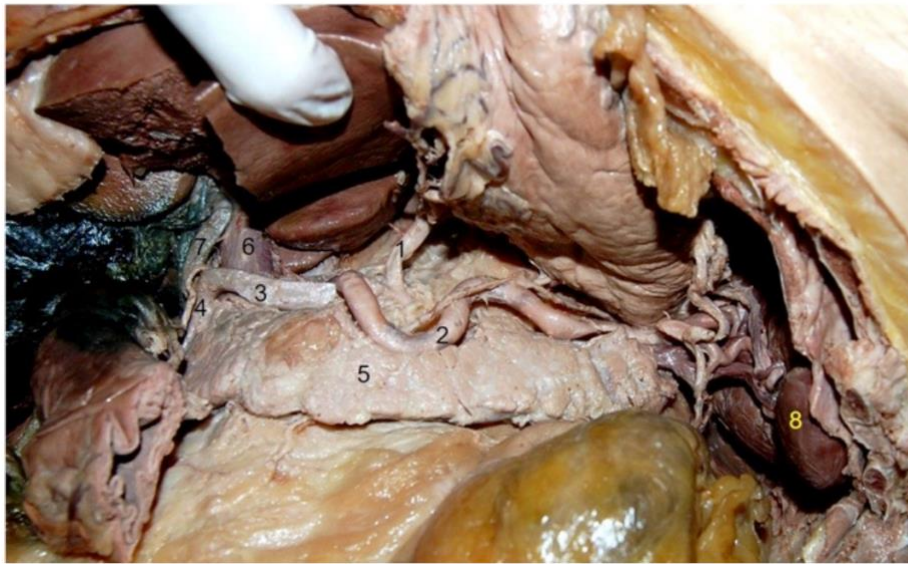
- 2 A. Transverse colon
- 4 B. Sigmoid colon
- 6 C. The mesentery
- 3 D. Descending colon
- 7 E. Small intestines
- 1 F. Ascending colon
- 5 G. Greater omentum



Which of the following peritoneal ligaments encloses the structures in the circumscribed area?

- A. Hepatogastric contains rt & lt gastric a; separates greater & lesser sac
- B. Splenorenal contains splenic a & v; posterolateral boundary of lesser sac
- C. Hepatoduodenal contains portal triad (proper name)
- D. Gastrosplenic contains short gastric a & v, left gastroepiploic a & v
- E. Gastrocolic contains gastroepiploic vessels

Term 2 Lab Exam Review



Match the names to the numbered structures

- 3 A. Common hepatic
- 7 B. Proper hepatic artery
- 5 C. Pancreas
- 1 D. Left gastric
- 2 E. Splenic
- 4 F. Gastroduodenal
- 8 G. Spleen
- 6 H. Portal vein



Jejunum: feathery appearance (deep mucosal folds)

